Toy Parameter Scan Plan for the Closed Minimal r_d Model

Operational scan plan • shared-parameter test for enlarged horizon and benchmark behavior

Purpose

This scan is designed to test whether one shared parameter region can simultaneously support an enlarged drag horizon and healthy benchmark behavior. The model now has a fixed phase-weight function, a fixed radiation-flow scalar, and a closed backreaction term, so the next step is disciplined exploration rather than further symbolic expansion.

Core scan parameters

Parameter	Role	Suggested v1 range
xi_0	transport-amplitude coupling	0.00 to 1.00
eta_T	energy-density backreaction amplitude	0.00 to 1.00
omega_0	preferred phase-alignment value	fit-centered or fixed reference
sigma_omega	phase-coherence width	0.05 to 2.00
beta_-	radiative transport weight	0.00 to 2.00
beta_+	geometry-coupled weight	0.00 to 2.00
beta_x	mixed-sector cooperation weight	0.00 to 2.00
w_T	effective equation-of-state of rho_T	test fixed cases: -1/3, 0, 1/3

Fixed v1 choices

Use Pi(omega) = 1 / (1 + ((omega - omega_0)/sigma_omega)^2) and LambdaHat_L = X/(1+X), with X = |L_mu L^mu| / L_*^2. For the first scan, either hold the background proxies for <F_-^2>, <F_+^2>, and |L_mu L^mu| fixed by epoch template, or test them with simple scaling laws rather than introducing a full new dynamical system.

Minimal observables to compute

1. r_d^RUT from the closed horizon integral.
2. Delta r_d / r_d relative to the baseline model.
3. The weighted drag-era averages <Delta_c>_d and <Delta_H>_d.
4. Whether the enlarged-horizon condition <Delta_c>_d > <Delta_H>_d holds.
5. Whether the same parameter region remains compatible with the broader benchmark lane you want to compare against.

Scan order

Stage 1: hold beta_- = beta_+ = 1 and beta_x = 0 to test pure separate-sector support.
Stage 2: turn on beta_x and scan mixed-sector cooperation.
Stage 3: vary sigma_omega to see whether the enlarged horizon prefers narrow or broad phase alignment.
Stage 4: vary eta_T against xi_0 to locate the transport-versus-backreaction boundary.

Stage 5: identify any shared region that both enlarges r_d and preserves acceptable benchmark behavior.

Key diagnostic plots

• r_d versus xi_0
• r_d versus eta_T
• contour map in (xi_0, eta_T)
• contour map in (omega_0, sigma_omega)
• contour map in (beta_x, sigma_omega)
• scatter of enlarged-horizon candidates colored by benchmark score

Decision rule

A promising region is one in which the model produces a stable positive shift in r_d without requiring excessive backreaction, and where that same region does not degrade the broader benchmark lane beyond acceptable tolerance. In compact form: seek regions with <Delta_c>_d > <Delta_H>_d and acceptable empirical carry-over.

Practical note

Do not inflate the scan too early. The first goal is not exhaustive realism; it is to find out whether the closed minimal model has a credible shared region at all. If it does, then the next step is to replace the epoch templates with more explicit field evolution.